

FAIRBANKS 11 NE, AK, USA

WMO: 702630

Lat: 64.9735N

Lon: 147.5100W

Elev: 348

StdP: 97.22

Time Zone: -9.00 (NAK)

Period: 02-19

WBAN: 26494

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.3	-30.0	-35.6	0.1	-32.0	-33.4	0.2	-29.4	6.1	-12.1	5.3	-11.9	0.1	N/A	0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	25.2	14.3	23.5	13.9	21.7	13.2	15.8	22.7	15.0	21.2	14.1	19.6	1.9	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.3	9.9	17.3	12.4	9.3	16.2	11.5	8.8	15.5	45.3	22.9	42.7	21.2	40.4	19.8	19.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.8	4.0	3.5	DB	-34.5	28.1	3.2	1.6	-36.8	29.3	-38.7	30.3	-40.5	31.2	-42.8	32.4	(4)
			WB	-33.9	17.5	3.2	1.1	-36.2	18.3	-38.0	18.9	-39.8	19.5	-42.1	20.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.0	-15.6	-11.0	-9.0	1.1	9.5	14.5	15.5	12.7	7.3	-1.0	-11.0	-13.1	(6)
(7)		DBStd	12.81	9.56	8.26	7.53	5.84	4.63	3.51	3.05	3.53	4.42	5.72	7.32	7.72	(7)
(8)		HDD10.0	4115	793	589	588	269	66	4	2	12	101	344	631	716	(8)
(9)		HDD18.3	6699	1051	822	847	518	276	124	97	178	331	601	881	974	(9)
(10)		CDD10.0	481	0	0	0	2	49	139	172	96	19	2	0	0	(10)
(11)		CDD18.3	23	0	0	0	0	1	9	9	4	0	0	0	0	(11)
(12)		CDH23.3	157	0	0	0	0	8	63	61	26	0	0	0	0	(12)
(13)		CDH26.7	14	0	0	0	0	0	7	5	2	0	0	0	0	(13)
(14)	Wind	WSAvg	1.3	0.9	1.1	1.5	1.7	1.6	1.5	1.4	1.4	1.2	0.9	0.8	(14)	
(15)	Precipitation	PrecAvg	323	15	13	9	9	17	44	71	56	38	19	17	17	(15)
(16)		PrecMax	472	64	41	54	70	53	78	147	95	77	38	46	76	(16)
(17)		PrecMin	220	3	3	0	0	4	15	5	9	8	6	2	1	(17)
(18)		PrecStd	60	14	10	10	14	11	18	37	22	17	8	14	14	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.6	6.7	8.9	17.3	24.4	27.7	27.4	25.8	19.3	16.4	6.1	5.7	(19)
(20)		MCWB	3.9	2.1	3.2	7.6	11.8	15.3	15.6	14.9	11.1	5.9	1.6	1.1	(20)	
(21)		2%	DB	3.6	4.0	5.8	13.5	21.7	25.2	25.0	23.6	17.3	10.8	3.8	3.0	(21)
(22)		MCWB	0.2	-0.1	1.1	5.7	10.9	13.9	15.3	14.4	10.1	5.2	0.3	-0.4	(22)	
(23)		5%	DB	1.2	2.3	3.7	11.2	19.5	23.1	23.4	20.7	15.6	8.7	1.6	0.7	(23)
(24)		MCWB	-1.3	-1.1	-0.4	4.5	10.0	13.4	14.8	13.3	9.2	3.9	-0.5	-1.9	(24)	
(25)		10%	DB	-2.1	0.3	1.5	9.2	17.4	21.3	21.6	18.7	13.9	6.7	-0.9	-1.7	(25)
(26)		MCWB	-3.9	-2.6	-1.6	3.4	8.9	12.6	14.1	12.4	8.3	2.6	-2.3	-3.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.8	1.4	3.9	7.3	12.9	16.9	17.4	16.0	11.8	6.6	1.6	1.3	(27)
(28)		MCDWB	7.4	3.6	9.3	13.6	22.0	24.4	23.8	22.2	17.2	11.7	3.7	4.6	(28)	
(29)		2%	WB	1.0	-0.3	2.1	5.2	11.8	15.6	16.1	14.6	10.6	5.5	0.1	0.1	(29)
(30)		MCDWB	3.5	3.2	6.1	10.9	20.8	22.8	23.1	19.6	15.5	10.0	2.5	3.2	(30)	
(31)		5%	WB	-0.8	-1.8	0.8	4.1	10.7	14.4	15.3	13.7	9.9	4.1	-1.3	-1.4	(31)
(32)		MCDWB	1.8	1.0	4.3	9.8	18.8	21.3	21.8	18.2	14.5	8.1	0.5	0.9	(32)	
(33)		10%	WB	-2.5	-3.2	-0.8	3.3	9.6	13.4	14.6	12.8	9.1	3.0	-3.0	-3.1	(33)
(34)		MCDWB	0.0	-0.8	2.4	8.5	16.8	19.6	20.3	17.0	13.3	6.2	-1.8	-1.2	(34)	
(35)	Mean Daily Temperature Range	MDBR	6.0	6.8	7.9	8.5	9.5	9.2	8.3	7.9	7.2	5.5	5.4	6.1	(35)	
(36)		5% DB	MCDBR	7.3	7.1	8.8	10.3	11.8	11.3	11.0	10.6	9.5	7.4	6.7	8.0	(36)
(37)		MCWBR	5.0	5.0	5.6	5.5	5.2	4.9	4.7	4.5	4.7	4.3	4.9	5.6	(37)	
(38)		5% WB	MCDBR	7.3	7.0	8.4	9.5	11.1	9.8	9.5	8.5	8.3	6.9	6.6	7.2	(38)
(39)		MCWBR	5.1	5.1	5.4	5.2	5.0	4.6	4.4	4.3	4.5	4.0	4.9	5.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.164	0.238	0.246	0.297	0.333	0.367	0.367	0.380	0.303	0.254	N/A	0.170	(40)	
(41)		taud	1.938	2.218	2.308	2.296	2.445	2.360	2.384	2.344	2.546	2.455	N/A	1.940	(41)	
(42)		Ebn at Noon	391	684	859	878	877	845	834	778	793	679	N/A	115	(42)	
(43)		Edh at Noon	22	52	75	99	95	106	101	96	63	42	N/A	7	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.15	0.84	2.65	4.59	5.24	5.50	4.76	3.62	2.28	0.94	0.23	0.04	(44)	
(45)		RadStd	0.01	0.07	0.16	0.29	0.39	0.46	0.34	0.36	0.27	0.11	0.03	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	+0.86	N/A	N/A	N/A	N/A	N/A	-326	-354	N/A	N/A

Nomenclature: See separate page